

Complete raw direct-future Doob split and the spatial one-use boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-31 · this version issued 2026-07-31 · v1.0

1. Purpose and scope; authority

R-136 recovered the R-125 future variance by common-heat replicas, proved the complete raw endpoint telescope in each replica, and selected a projected raw sequential one-use theorem as the next child. Its displayed production interface separated a direct prefix from a future-feedback connection, but it did not prove the full index decomposition or assign the connection to the physical insertion shells that created it.

This note closes that abstract indexing point. Give each physical insertion shell k its chronological insertion rank $t(k)$. A complete raw sequential telescope has two triangles at every forward reveal root: the direct triangle $t(k) \leq r$ and the future-feedback triangle $t(k) > r$. On the canonical aligned whole-shell chart $t(k) = k$, which is the only setting where the shorter $k \leq r$ notation is used below. The second triangle is not optional. Its six-row current coordinate is exactly the R-079 connection. Trace curvature, forest, low, rational, and Cartan companions remain separate complete-owner obligations. Consequently a q_k ledger for only $t(k) \leq r$ is incomplete. The future current triangle must either be transposed to its last insertion, as R-102 does in a narrower regular rational class, or remain unsplit inside a complete signed owner with a direct integrated estimate.

The note also places the inherited R-084 Ornstein–Uhlenbeck identity at the correct raw coordinate. It removes the need to estimate an explicit Malliavin derivative of terminal feedback: the OU integral is exactly the squared forward-Doob increment. This is a derivative-free reduction, not a new spatial estimate and not a new claim of the R-084 theorem. An exact Fourier modulation reuses NG-2026-07-31-A13-BOUNDED-MULTIPLIER-TO-SHELL-DECAY to show that OU, replicas, and bounded coefficients do not create the missing $2^{-4(m-k)/3}$ decay. A finite Gram lemma likewise shows that the bookkeeping direct sums do not make distinct physical insertion columns orthogonal.

The binding authorities are R-063, R-079, R-084, R-087, R-088, R-102, R-104, R-123, R-125, R-131, R-134, R-135, and R-136. R-137 claims novelty only for the exact chronologically typed all-insertion split of the R-136 raw telescope and the corrected successor interface. It is not a new spatial or Carleson estimate. It proves no production raw spatial estimate, no once-owned q_k sum, no signed forest-current lower bound, no positive collar headroom, neither A13 gate, no Nelson theorem, and no Sector-A closure.

2. Complete owner and index discipline

Let o be a complete owner after every same-root visit has telescoped and every contraction-closed exhaustive output cluster has been placed in its orthogonal direct sum. Write $\tau(o)$ for its retained covariance root, $r > \tau(o)$ for a forward reveal root, k for a physical control insertion shell, $t(k)$ for its chronological insertion rank, m for a spatial output shell, and $q \in \{f, f'\}$ for the two legal future replicas. The heat dummy is common to the two replicas. Define

$$\mathcal{K}_{\leq r} = \{k : k_0 < k \leq K, t(k) \leq r\}, \quad \mathcal{K}_{> r} = \{k : k_0 < k \leq K, t(k) > r\}. \quad (2.1)$$

These indices have different jobs:

$$\boxed{\text{owner } o \neq \text{reveal } r \neq \text{physical insertion } k \neq \text{output } m.} \quad (2.2)$$

Visits telescope before owner formation. Output clusters are orthogonal only after exhaustive recombination. Owners may be put in an external Hilbert direct sum for bookkeeping. None of these operations makes distinct k columns physically orthogonal.

For reference, R-123 fixes the six production rows through

$$q_A(w) = \frac{w^T S_A w}{|w|^2 + e}, \quad (2.3)$$

$$P_A(w; z) = 2(S_A w)^T z, \quad L_A(w; z) = 2\{(S_A - \alpha q_A(w)I)w\}^T z, \quad (2.4)$$

with $\alpha = 5/9$, $P = 4 + 10^{-12}$, $c_0 = 3/(250P)$, and $c_1 = 243/(8000P)$. Put

$$C_6(w)z = (\sqrt{c_0}P_A(w; z), \sqrt{c_1}L_A(w; z))_{A=1}^3, \quad B(w) = C_6(w)^* C_6(w). \quad (2.5)$$

The unaveraged current is $J = C_6(W)V$. All six rows stay in one vector. The trace and R-063 forest are scalar companions of this same owner, not additional copies of the current square.

3. Exact all-insertion raw Doob split

Fix one complete owner and suppress the replica superscript. R-136 gives the pathwise raw sequential telescope

$$X_o^{(K)} = Y_{o,\text{low}} + \sum_{k_0 < k \leq K} Y_{o,k}, \quad (3.1)$$

where

$$Y_{o,\text{low}} = X_o[A^{(k_0)}], \quad Y_{o,k} = X_o[A^{(k)}] - X_o[A^{(k-1)}]. \quad (3.2)$$

Every later feedback term is recomputed at the endpoint that appears in (3.2). Let $P_r = \mathbb{E}[\cdot \mid \mathcal{F}_r]$ and $d_r = P_r - P_{r-1}$.

Theorem 3.1 (complete raw reveal triangle)

For every finite temporally faithful chart and reveal root r ,

$$\boxed{d_r X_o^{(K)} = d_r Y_{o,\text{low}} + \sum_{k \in \mathcal{K}_{\leq r}} d_r Y_{o,k} + \sum_{k \in \mathcal{K}_{> r}} d_r Y_{o,k}.} \quad (3.3)$$

Separately, if every displayed $P_r Y_{o,\text{low}}$, $P_r Y_{o,k}$, and $P_r \sum_{k \in \mathcal{K}_{> r}} Y_{o,k}$ belongs to the root Sobolev domain, linearity gives

$$\boxed{D_r P_r X_o^{(K)} = D_r P_r Y_{o,\text{low}} + \sum_{k \in \mathcal{K}_{\leq r}} D_r P_r Y_{o,k} + D_r P_r \sum_{k \in \mathcal{K}_{> r}} Y_{o,k}.} \quad (3.4)$$

Proof. Apply the linear map d_r to the finite telescope (3.1), then partition the finite insertion set by $t(k) \leq r$ and $t(k) > r$. Under the separately stated termwise Sobolev hypotheses, apply $D_r P_r$ to that finite identity. No derivative is commuted through a random coefficient and no future control is frozen.

□

The two sums in (3.3) are the direct and future triangles. The future triangle can be nonzero although every $Y_{o,k}$ with $t(k) > r$ is inserted later in chronological insertion order: later controls may depend on root r . Thus strict-past adaptedness does not delete it.

4. The future triangle is the R-079 connection

Let $A^{[\leq r]}$ denote the same base/low datum as $A^{(K)}$ together with exactly the control insertions a_k on the common chart clock for which $t(k) \leq r$. Put

$$R_r^f = A^{(K)} - A^{[\leq r]} = \sum_{k \in \mathcal{K}_{>r}} a_k, \quad W_* = W_r + R_r^f, \quad V_* = V_r + DR_r^f. \quad (4.1)$$

The complete current difference has the exact two-channel form

$$\boxed{J_* - J_r = [C_6(W_*) - C_6(W_r)]V_r + C_6(W_*)DR_r^f.} \quad (4.2)$$

The first term is the future coefficient channel and the second is the future derivative-feedback channel. Since

$$D_r R_r^f = \sum_{k \in \mathcal{K}_{>r}} D_r a_k \quad (4.3)$$

need not vanish. On a bounded smooth cylindrical chart where the displayed mixed derivatives exist and commute, differentiation of (4.2) retains all four descendants

$$D_r(\Delta C V_r) = (D_r \Delta C)V_r + \Delta C D_r V_r, \quad (4.4)$$

$$D_r(C_* DR_r^f) = (D_r C_*)DR_r^f + C_* D(D_r R_r^f). \quad (4.5)$$

The current identity (4.2) does not by itself prove the trace or forest companions. The separate Gram/trace endpoint identity is

$$B_* - B_r = C_r^* \Delta C + \Delta C^* C_r + \Delta C^* \Delta C. \quad (4.6)$$

The curvature square $\Delta C^* \Delta C$ and the direct-future cross occur once in that trace companion. Forest and other signed-owner terms still need their own complete ownership. Outside the smooth-cylinder hypotheses, R-137 uses only the derivative-free Doob identity (5.1), not (4.4)–(4.5).

R-079 writes the same fact in root-current coordinates. With

$$f_r = d_r(J_r - J_0), \quad i_r = d_r(J_* - J_r), \quad (4.7)$$

one has

$$\begin{aligned} & \langle d_r J_0, f_r + i_r \rangle + \frac{1}{2} \|f_r + i_r\|^2 \\ &= \langle d_r J_0, f_r \rangle + \frac{1}{2} \|f_r\|^2 + \langle d_r J_r, i_r \rangle + \frac{1}{2} \|i_r\|^2. \end{aligned} \quad (4.8)$$

The cross $\langle f_r, i_r \rangle$ is inside the third term on the right. For the exact two-root fixture

$$J_0 = 0, \quad J_1 = \xi_1, \quad J_* = 2\xi_1 + \xi_2, \quad (4.9)$$

the physical-prefix energy is two, the complete terminal Doob energy is five, and the missing cross-plus-square correction is three. Therefore a prefix-only raw ledger deletes a load-bearing owner contribution.

5. Derivative-free OU reduction at the raw coordinate

Let $M_{o,r} = P_r X_o^{(K)}$ and $\Delta_{o,r} = d_r X_o^{(K)}$. If root r is a whitened finite Gaussian and $M_{o,r}$ lies in its Sobolev domain, the R-084 identity gives, for every deterministic spatial projection Π_m ,

$$\boxed{\mathbb{E} \|\Pi_m \Delta_{o,r}\|^2 = 2 \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m P_t^{(r)} D_r M_{o,r}\|_{\text{HS}}^2 dt.} \quad (5.1)$$

On Gaussian chaos $n \geq 1$, the right side is

$$2 \int_0^\infty e^{-2t} n e^{-2(n-1)t} dt = 1, \quad (5.2)$$

times the squared chaos norm.

Equation (5.1) is useful because its left side contains no explicit Malliavin derivative of a terminal adapted control. The complete feedback is retained inside the forward-Doob increment. This is the legal derivative-free form of the raw target. It does not permit the pointwise substitution of A^* into R-084's predictable three-channel formula, and it does not generate spatial decay: it reproduces the projected variance exactly.

Combining (3.3) and (5.1), the production problem is now precisely to bound the projected complete direct-plus-future raw triangle. Replicas give the same variance with their factor one half, but likewise supply no dyadic gain.

6. Conditional direct three-channel formula

For $t(k) \leq r$ (written $k \leq r$ only on the aligned chart), let F_a be one of the six coefficient rows, put $B = W^{(k-1)}$, and write $z = D_r B$. If the relevant $D_r P_r = P_r D_r$ commutation is certified on the displayed product, strict past gives $D_r a_k = 0$ and the direct atom is

$$\begin{aligned} \mathbf{t}_{a,r,k}^{\text{raw}}[z] = & \{DF_a(B + a_k) - DF_a(B)\}[z]^T DB \\ & + DF_a(B + a_k)[z]^T D a_k \\ & + \{F_a(B + a_k) - F_a(B)\}^T D z. \end{aligned} \quad (6.1)$$

These are exactly the coefficient, control-derivative, and root-derivative channels of R-088/R-135. The difference is load bearing: B in (6.1) is raw, while R-088 has already applied deterministic future heat. If the commutation is not proved, $D_r P_r Y_{o,k}$ remains the exact definition and (6.1) is not asserted.

For $t(k) > r$, formula (6.1) is not the full atom. The four descendants (4.4)–(4.5), the trace curvature (4.6), and the forest/current companions must be assigned by a finite-Fubini last-insertion theorem or kept in one unsplit signed connection owner. R-102 supplies this transposition only for its regular mutually orthogonal no-revisit rational class.

7. OU and bounded multipliers do not create spatial decay

For the direct triangle $t(k) \leq r$, the missing estimate at $s = 2/3$ would have the form

$$2 \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m P_t^{(r)} \widehat{\mathcal{T}}_{r,k}\|_{\text{HS}}^2 dt \leq 2^{-\frac{4}{3}(m-k)} q_k, \quad t(k) \leq r, \quad m \geq r + 5. \quad (7.1)$$

The OU identity cannot prove the exponent. On the complexification of the one-dimensional torus with normalized Fourier basis, embedded into the production torus, take

$$Y_{r,m}(\xi, x) = \xi e^{i2^r x} \cos((2^m - 2^r)x). \quad (7.2)$$

The multiplier has supremum one and $d_r Y_{r,m} = Y_{r,m}$. The coefficient at frequency 2^m is exactly $1/2$. For this diagnostic let Π_m^{sharp} be the rank-one projection onto that normalized carrier; its squared norm is $1/4$. For $k = r$ and $m - r = 6$, the sharp-projector analogue of (7.1) forces

$$q_r \geq \frac{1}{4} 2^{\frac{4}{3}6} = 64. \quad (7.3)$$

Moving $m - r$ outward makes the required constant diverge. This does not identify Π_m^{sharp} with every smooth production projector.

This is the already registered bounded-multiplier inference no-go, now placed at the raw Doob coordinate. It is not a production-current counterexample: (7.2) does not satisfy the production order-two principal support cancellation or the R-134 joint spatial data. Its consequence is methodological and sharp. A valid proof must show that every complete raw direct channel has the R-087 Q_r -supported three-channel paracomposition structure with principal annihilation or explicitly proved R-134-type joint spatial data. A future column must additionally be transposed to a last insertion with its own $m \geq k + 5$ geometry, or be controlled by a direct integrated signed estimate.

8. Bookkeeping direct sums are not a k -Bessel theorem

Let $T_1, \dots, T_N$ be columns in a Hilbert space and let $G_{k\ell} = \langle T_k, T_\ell \rangle$. A deterministic diagonal estimate

$$\left\| \sum_k c_k T_k \right\|^2 \leq \sum_k q_k |c_k|^2 \quad \text{for every } c \quad (8.1)$$

holds exactly when

$$\boxed{G \preceq \text{diag}(q_1, \dots, q_N)}. \quad (8.2)$$

This is the finite Gram/Loewner criterion.

For coherent unit columns $T_k = u$, G is the all-ones matrix. If every $q_k > 0$ and $D = \text{diag}(q)$ majorizes G , then $\sum_k q_k^{-1} \leq 1$. Cauchy–Schwarz gives

$$\sum_k q_k \geq N^2, \quad (8.3)$$

with equality at $q_k = N$. The sum of the individual energies is only N . Thus bookkeeping cannot turn coherent physical columns into a constant-one Bessel family. With two opposite columns, the physical sum is zero while the sum of separated squares is two, so unjustified splitting can also destroy cancellation.

R-135 does not assume cross- k orthogonality; it pays cross terms by weighted Cauchy–Schwarz after each individual atom estimate is established. The point of (8.2)–(8.3) is narrower: owner, cluster, visit, or reveal direct sums cannot be cited as a proof of the missing deterministic $\sum_k q_k \leq Q$. Same- k coherent pieces must first be recombined, and the remaining physical k family needs a production Carleson/Bessel estimate or the R-135 weighted route with a separately proved one-use q sum.

9. Exact relation to the R-135 post-heat theorem

R-135 proves the post-heat R-088 direct atom estimate at

$$\alpha = \frac{2}{5}, \quad s = \frac{2}{3}, \quad (9.1)$$

with exponent margins

$$\frac{1}{15}, \quad \frac{4}{15}, \quad \frac{16}{15}. \quad (9.2)$$

Writing its requested leading factor two into the charge gives

$$q_k = 2q_k^{\text{seq}} = 2C_e 2^{-7k/5} W_k^{\text{seq}}. \quad (9.3)$$

This is a rigorous direct-triangle theorem only after future heat. It does not estimate (4.2), assign the $k > r$ triangle, or prove $\sum_k q_k < \infty$. If a complete raw theorem establishes (7.1), transposes the future

triangle to R-135-compatible last-insertion atoms, and proves the one-use sum, R-135 then supplies the nonorthogonal $\gamma = 7/12$ square constant

$$0.658881625872614\dots \quad (9.4)$$

No part of R-137 invokes that conditional conclusion prematurely.

10. Proof, failure, and successor map

Route	Verdict	Reusable conclusion
Complete raw insertion telescope	proved	Both $t(k) \leq r$ and $t(k) > r$ triangles occur in (3.3).
Future triangle deletion	rejected	R-079 gives cross plus square correction three in the exact fixture.
Raw OU coordinate	exact, reused	(5.1) replaces explicit feedback derivatives by the projected Doob increment with constant one.
OU as spatial gain	failed inference	The modulation fixture forces $q_r \geq 64$ already at gap six.
Bookkeeping as k orthogonality	failed inference	Coherent columns require diagonal trace N^2 , not N .
R-087/R-135 direct transfer	post-heat only	Margins (9.2) survive, but the raw connection is absent.
R-102 last insertion	conditional template	It closes only the declared regular rational no-revisit class.
Production raw one-use	open	Owner identification, both triangles, spatial structure, and $\sum q_k \leq Q$ remain to prove.

The proof-search records are EXP-000540–EXP-000545. They separately record the all-insertion split, the derivative-free OU reduction, the reused spatial no-go, the Gram/Bessel boundary, the corrected successor, and the independent audit repair of the assertion accounting, chronological notation, Sobolev hypothesis, and derived-summary propagation. The integrated verifier contains the child rows: its 263 unique package assertions comprise 114 embedded child assertions and 149 integrator-only assertions, so the child totals must not be added a second time. The negative registry authorities are the existing NG-2026-07-25-A13-ROOT-ORTHOGONALITY-ONE-USE and NG-2026-07-31-A13-BOUNDED-MULTIPLIER-TO-SHELL-DECAY; no duplicate formal negative is created.

11. Devil’s-advocate review

1. **Objection: The $t(k) > r$ terms vanish because they are future. DISMISSED.** They are future in insertion order, not independent of root r . Equation (4.3) is generally nonzero, and the R-079 fixture has $f_1 = i_1 = \xi_1$.
2. **Objection: R-084 already proves R-137. VALID provenance correction, with a new consequence.** R-084 owns the OU identity and its predictable-root formula. R-137 does not reclaim them. Its new statement is the exact all-insertion split of the R-136 raw telescope and the conclusion that the future triangle must be assigned in the successor.
3. **Objection: Applying D_r to (3.1) differentiates an arbitrary terminal control. DISMISSED within the stated Sobolev domain.** Equation (3.3), the primary derivative-free statement, uses only conditional expectations. Equation (3.4) is asserted only when every displayed $P_r Y_{o,\text{low}}$, $P_r Y_{o,k}$, and $P_r \sum_{k \in \mathcal{K}_{>r}} Y_{o,k}$ belongs to the root Sobolev domain, exactly as required in Theorem 3.1. No termwise derivative estimate is used to claim spatial decay.

4. **Objection: The modulation fixture is a production counterexample. DISMISSED.** It tests bounded multipliers alone and omits the production principal-support cancellation, heat, trace, forest, and signed owner. It is an inference no-go only.
5. **Objection: Owners and clusters are direct sums, so the physical k columns are orthogonal. DISMISSED.** Those direct sums have different indices. The coherent Gram family (8.3) gives the exact cost of confusing them.
6. **Objection: Equation (9.3) proves the raw estimate. UPHELD as an overclaim.** It is the already proved post-heat direct atom. The raw future connection, owner realization, and one-use sum remain open.
7. **Objection: The all-insertion identity closes controlled-shell one-use. UPHELD as an overclaim.** It repairs the indexing and prevents a missing channel. It proves no production spatial decay or positive signed form.

External review is invited for the split point in (3.3), the distinction between (3.3) and (3.4), both channels in (4.2), the OU factor two in (5.1), the Fourier coefficient and exponent in (7.3), the Gram criterion (8.2), and the no-overclaim boundary in Section 12.

12. Reproduction, exact successor, and Result footer

Run from the repository root:

```
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
& $py codes/foundations/
  a13_classii_raw_doob_triangle_spatial_carleson_boundary.py
& $py codes/foundations/
  a13_classii_raw_doob_triangle_spatial_carleson_boundary_independent.py
& $py codes/foundations/
  a13_classii_raw_doob_triangle_spatial_carleson_boundary_verify.py
```

The primary executable checks 60 exact assertions. The non-importing standard-library audit checks 54 independently assembled assertions. The integrated verifier pins the note, negative and exploration records, authorities, generated surfaces, and scope firewalls.

The corrected successor is

$$\boxed{\text{A13-CLASSII-COMPLETE-RAW-DOOB-DIRECT-FUTURE-SPATIAL-ONE-USE-BOUND.}} \quad (12.1)$$

It must:

1. instantiate (3.3) in the actual R-123/R-125 owner with complete low, visits, exhaustive clusters, all six rows, trace, rational and Cartan companions, R-063 forest, and common heat exactly once;
2. prove the R-087-type spatial structure for the $t(k) \leq r$ direct triangle;
3. either transpose the $t(k) > r$ connection to unique last-insertion atoms with $m \geq k + 5$ and R-135-compatible charges, or prove a direct integrated signed substitute, without duplicating the R-079 cross, square, or trace curvature;
4. prove the appropriate direct and transposed spatial estimates and a deterministic nonnegative $\sum_k q_k \leq Q$ without owner, reveal, visit, cluster, row, or refinement multiplicity;
5. keep the balanced/near current and R-063 forest in one signed form; then use R-135 once only in the compatible transposed branch, or the proved integrated substitute otherwise, and test one finite collar's headroom.

Result ID: A13-CLASSII-RAW-DOOB-TRIANGLE-SPATIAL-CARLESON-
BOUNDARY

Ledger ID: R-137

Precise statement: The complete raw sequential telescope has both direct $t(k) \leq r$ and future-feedback $t(k) > r$ triangles at every forward reveal root. Its six-row future current is the R-079 two-channel connection and cannot be deleted; trace, forest, and other signed-owner companions remain separate. R-084's OU identity gives an exact derivative-free projected Doob norm but no spatial decay. A bounded multiplier cannot imply the dyadic exponent, and bookkeeping direct sums do not orthogonalize physical k columns.

Scope: fixed finite cutoff and positive floor; one finite temporally faithful strict-past chart with an explicit chronological insertion rank $t(k)$; square-integrable complete owners; recombined visits; exhaustive clusters; common heat; root Sobolev statements only where explicitly declared.

Dependencies: R-063, R-079, R-084, R-087, R-088, R-102, R-104, R-123, R-125, R-131, R-134, R-135, and R-136, with the two existing negative-result authorities named in Section 10.

Evidence grade: EXACT, EXECUTED.

Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/
a13_classii_raw_dooob_triangle_spatial_carleson_boundary_verify.py

Expected output: R-137 integrated PASS with primary 60/60, independent 54/54, and the manifest-pinned integrated assertion contract.

Falsification gate: failure of the all-insertion telescope, either R-079 connection channel, the factor-three fixture, OU factor two, Fourier coefficient one half, gap-six q floor 64, coherent Gram trace N^2 , any authority or scope pin, or any executable/PDF/release contract.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no production owner realization, raw spatial decay, future-triangle last-insertion estimate, one-use q sum, signed forest-current lower bound, headroom, low/matching/gap/anchor data, A13 gate, OVERLAP_src, Nelson, measure, removal limit, or Sector-A closure is proved.

Next required action: prove

A13-CLASSII-COMPLETE-RAW-DOOB-DIRECT-FUTURE-SPATIAL-ONE-USE-BOUND with a production owner realization, both raw triangles, a spatial estimate, and a deterministic once-owned q_k sum before testing collar headroom.

Proof complete: false. Sector A closed: false.